4CS015 Fundamentals of Computing – Workshop #6

TO BE SUBMITTED AS PART OF YOUR PORTFOLIO

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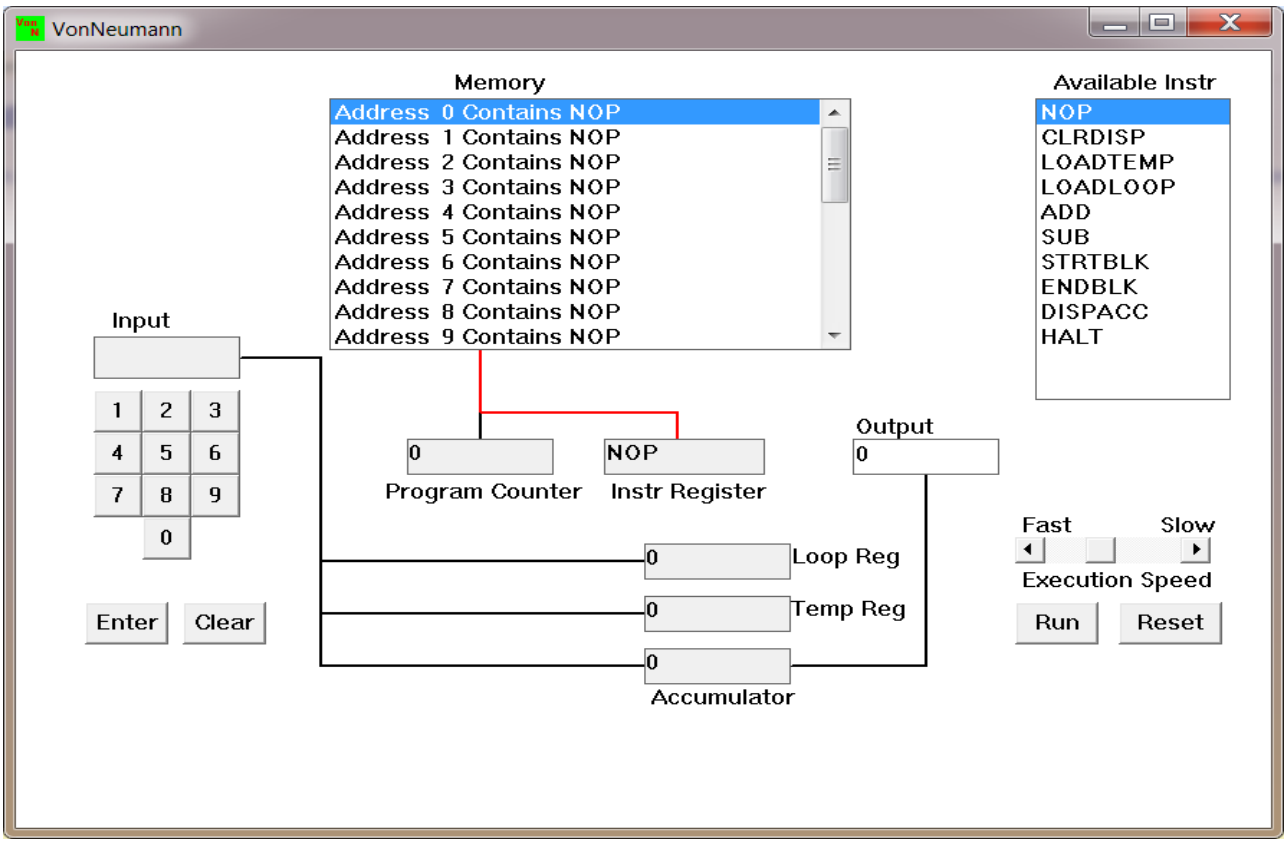
This is a marked workshop. You will need to complete the workshop and then submit a copy of this document with a title that follows the following format ("YOURNAME 1234567 wsp6.docx"), via CANVAS, by the deadline. The Von Neumann Simulator can be found on Canvas.

Workshop tasks:

1. Von Neumann-Simulator. This program simulates an elementary computer with the von Neumann architecture.

   Download the von Neumann Simulator (VonNeumann.exe) program from Canvas in the Week 7 topic folder. Save it in your Documents folder and run it. You will see a window like this:



   The simulator has a small program memory area which is available for programming. To enter your program instructions simply click on the "Available" instruction on the list on the

right and then click on the “Memory” location you wish to populate.

The simulator understands only the following ten instructions:

NOP	No Operation, i.e. do nothing.
LOADTEMP	Get a number from the keypad, completed by the Enter key, into the Temporary Register.
LOADLOOP	Get a number from the keypad, completed by the Enter key, into the Loop Register.
CLRDISP	Clear the Display.
ADD	Add the Temporary Register to the Accumulator
SUB	Subtract the Temporary Register from the Accumulator
DISPACC	Display the contents of the Accumulator
STRTBLK	Start of Loop Block
ENDBLK	End of Loop Block
HALT	Halt. Stop Program

Load the following program into the memory:
LOADTEMP
ADD
DISPACC
HALT

To do this, first click on the “LOADTEMP” in the list of instructions on the right of simulator window. Then click on Memory location with “Address 0 Contains NOP”. This will then change into “Address 0 Contains LOADTEMP”. Repeat the process with “Address 1” and so on until the whole program is loaded.

Run the program by clicking on the “Run” button. The simulator would highlight the Address 0 location and then pause. It is executing the instruction “LOADTEMP” which requires you to input a number into the keypad.

Click 2 or 3 numbers on the keypad and then click the “Enter” button. The simulator will then resume running the program and execute the instruction “ADD”. This adds the number that you just entered, to the zero in the accumulator.

The next instruction is “DISPACC” which stands for “Display Accumulator”, and it does

exactly that. After than the simulator stops running the program when it executes the instruction “HALT”.

a. Load the following program into the simulator:

LOADTEMP

ADD

LOADTEMP

ADD

DISPACC

HALT

What do you think it does? Write your answer below (10 marks) (Make sure you include details of what is happening at every program pointer position to get full marks, you can include screenshots as well as descriptions to help illustrate your answer)

<u>Ans</u>. The above program simply loads two value one by and add them together and displays their sum. It uses a temporary register (TEMP) and an accumulator (ACC) where each LOADTEMP loads value to the accumulator. Each ADD adds the loaded value to the current accumulator value. DISPACC displays the result and HALT stops the program. Here, accumulator starts at 0 and TEMP holds some input value and program pointer moves to the next step after each execution.

<u>Program pointer 1: LOADTEMP</u>

Here the value that is entered is stored to the TEMP. Program counter is 0.

TEMP = 12
ACC = 0
Output = 0

Program pointer 2: ADD

Here the entered value is added to the current value of accumulator. Program counter is 1. This part gets executed automatically after the user inputs the first value.

ACC = 0 + 12 = 12
Output = 0

Program pointer 3: LOADTEMP

Here the value that is entered is stored to the TEMP. Program counter is 2.

TEMP = 13
ACC = 12
Output = 0

Program pointer 4: ADD

Here the entered value is added to the current value of accumulator. Program counter is 3. This part gets executed automatically after the user inputs the second value.

ACC = 12 + 13 = 25
Output = 0

Program pointer 5: DISPACC

Here the current value of accumulator (ACC) is displayed to the output.

TEMP = 13
ACC = 25
Output = ACC = 25

b. Write a program to add 3 different numbers together. List your program below (10 marks)

The are two program to perform this task:

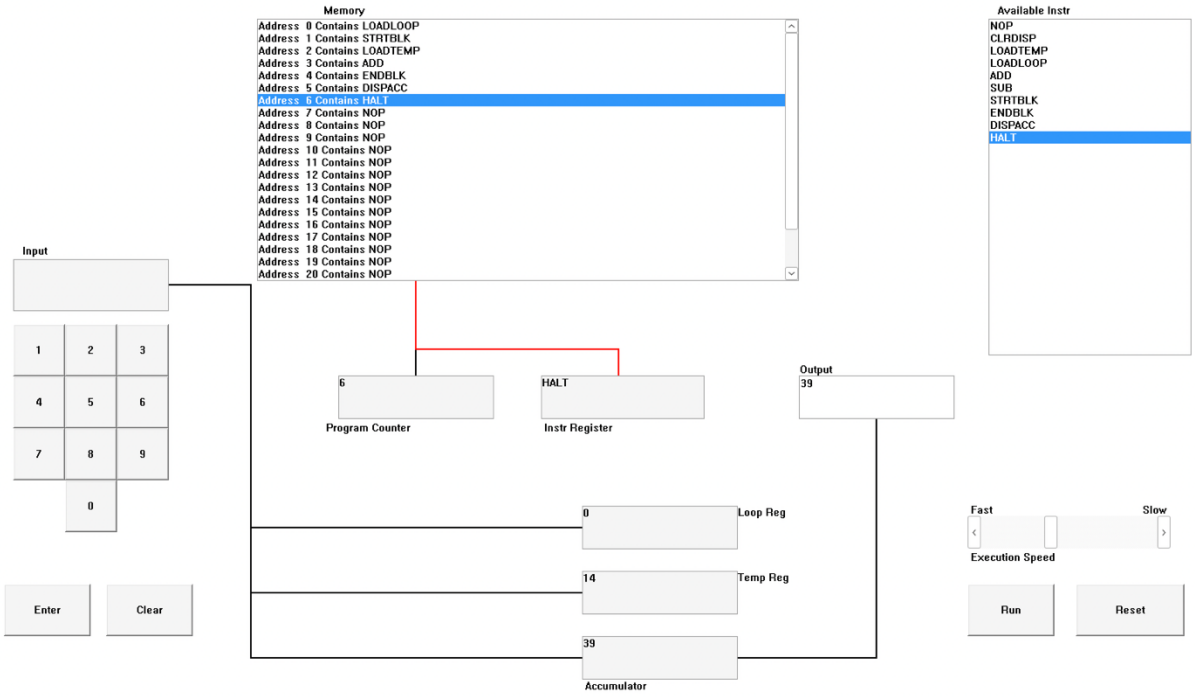
1st Program

```
LOADLOOP    (3)
STRTBLK
LOADTEMP
ADD
ENDBLK
DISPACC
HALT
```

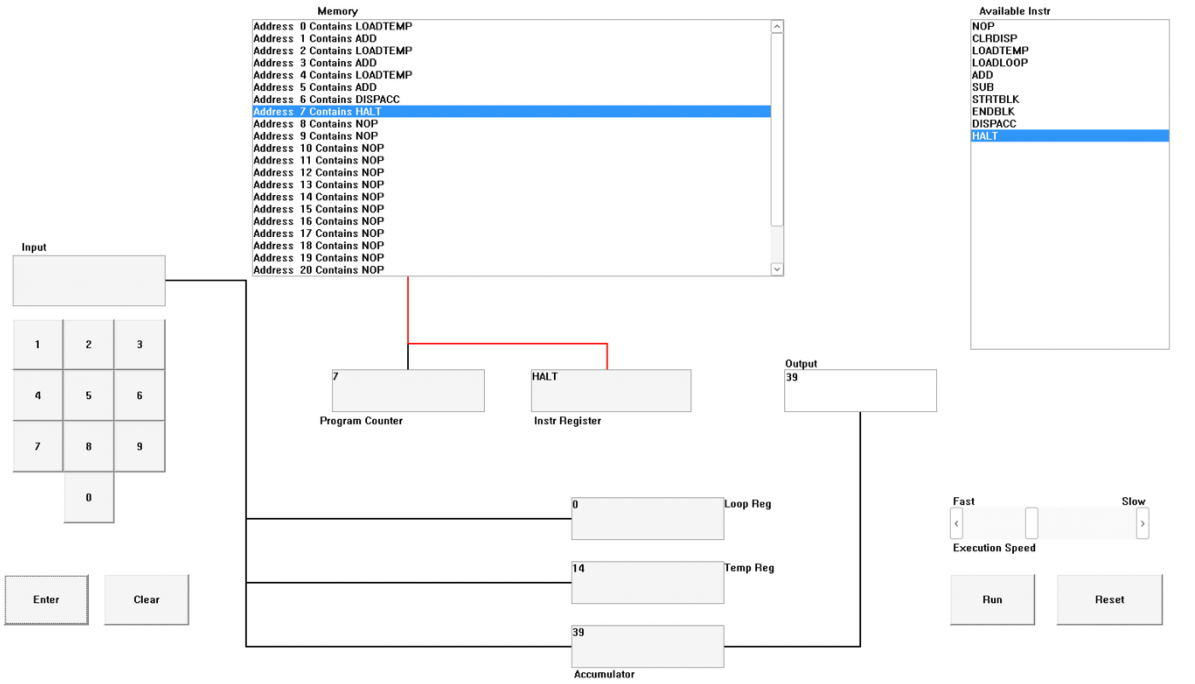
2nd Program

```
LOADTEMP
ADD
LOADTEMP
ADD
LOADTEMP
ADD
DISPACC
HALT
```

LOADLOOP value must be 3



2nd program



c. **Write a program to subtract a number from another number. List your program below (10 marks)**

LOADTEMP
ADD
LOADTEMP
SUB
DISPACC
HALT

<u>12-2 = 10</u>

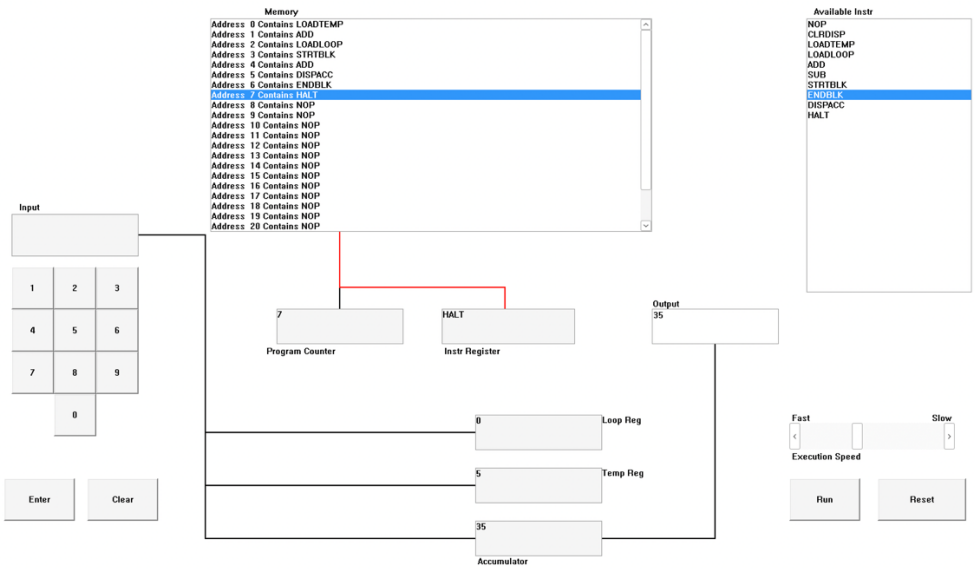
d. **Load the following program into the simulator:**

LOADTEMP
ADD
LOADLOOP
STRTBLK
ADD
DISPACC
ENDBLK
HALT

Run it and when it reaches the LOADTEMP instruction, enter 5 on the keypad and click the "Enter" button. When it reaches the LOADLOOP instruction, enter 6. What do you think the program does? Write your answer below in the form of an equation (10 marks) (Make sure you include details of what is happening at every program pointer position to get full marks, you can include screen shots as well as descriptions to help illustrate your answer)

<u>Ans.</u> The above program performs the equation: **5 + (6 X 5) = 35**. When user enter a value 5 the programs run a loop and add 5 to the current value 6 times in the loop and displays the result 35 at the end. In the loop 5 is added 6 times to the current value i.e. 5+5+5+5+5+5.

LOADTEMP = 5
LOADLOOP = 6



Temporary register = TEMP
Accumulator = ACC

<u>Program pointer 1: LOADTEMP</u>

The program starts and loads the entered value to the TEMP. program counter is 0.

TEMP = 5
ACC = 0 (Accumulator starts at 0)

<u>Program pointer 2: ADD</u>

Here the value in TEMP is added to the current value of ACC. Program counter is 1. This part gets executed automatically after the value is entered.

TEMP = 5
ACC = 0 + 5 = 5

<u>Program pointer 3: LOADLOOP</u>

Here the programs load the entered value 6 that means the loop block will run 6 times. Program counter is 2. Loop register stores the entered the value.

TEMP = 5
ACC = 5
Loop Reg = 6

<u>Program pointer 4: STRTBLK</u>

Here the block of loop is started. Program counter is 3.

TEMP = 5
ACC = 5
Loop Reg = 6

<u>Program pointer 5: ADD</u>

Here the value in TEMP is added to the current value of ACC. Program counter is 4. This part gets executed automatically after the loop value is entered.

TEMP = 5
ACC = 5 + 5 = 10
Loop Reg = 6

<u>Program pointer 6: DISPACC</u>

Here the value in accumulator is displayed in the output. Program counter is 5.

TEMP = 5
ACC = 10
Output = 10 Loop Reg = 6

Program pointer 7: ENDBLK

This block ends the loop when value in loop register becomes 0 or passes the program counter to 4 if loop register isn't 0. Program counter is 6.

TEMP = 5
ACC = 10
Output = 10 Loop Reg = 5

Program pointer 8: ADD

Here the value in TEMP is added to the current value of ACC. Program counter is 4. This part gets executed automatically after the loop value is entered.

TEMP = 5
ACC = 10 + 5 = 15
Loop Reg = 5

Program pointer 9: DISPACC

Here the value in accumulator is displayed in the output. Program counter is 5.

TEMP = 5
ACC = 15
Output = 15 Loop Reg = 5

Program pointer 10: ENDBLK

This block ends the loop when value in loop register becomes 0 or passes the program counter to 4 if loop register isn't 0. Program counter is 6.

TEMP = 5
ACC = 15
Output = 15 Loop Reg = 4

Program pointer 11: ADD

Here the value in TEMP is added to the current value of ACC. Program counter is 4. This part gets executed automatically after the loop value is entered.

TEMP = 5
ACC = 15 + 5 = 20
Loop Reg = 4

Program pointer 12: DISPACC

Here the value in accumulator is displayed in the output. Program counter is 5.

TEMP = 5
ACC = 20
Output = 20 Loop Reg = 4

Program pointer 13: ENDBLK

This block ends the loop when value in loop register becomes 0 or passes the program counter to 4 if loop register isn't 0. Program counter is 6.

TEMP = 5
ACC = 20
Output = 20 Loop Reg = 3

Program pointer 14: ADD

Here the value in TEMP is added to the current value of ACC. Program counter is 4. This part gets executed automatically after the loop value is entered.

TEMP = 5
ACC = 20 + 5 = 25
Loop Reg = 3

Program pointer 15: DISPACC

Here the value in accumulator is displayed in the output. Program counter is 5.

TEMP = 5
ACC = 25
Output = 25 Loop Reg = 3

Program pointer 16: ENDBLK

This block ends the loop when value in loop register becomes 0 or passes the program counter to 4 if loop register isn't 0. Program counter is 6.

TEMP = 5
ACC = 25
Output = 25 Loop Reg = 2

Program pointer 17: ADD

Here the value in TEMP is added to the current value of ACC. Program counter is 4. This part gets executed automatically after the loop value is entered.

TEMP = 5
ACC = 25 + 5 = 30
Loop Reg = 2

Program pointer 18: DISPACC

Here the value in accumulator is displayed in the output. Program counter is 5.

TEMP = 5
ACC = 30
Output = 30 Loop Reg = 2

Program pointer 19: ENDBLK

This block ends the loop when value in loop register becomes 0 or passes the program counter to 4 if loop register isn't 0. Program counter is 6.

TEMP = 5
ACC = 30
Output = 30 Loop Reg = 1

Program pointer 17: ADD

Here the value in TEMP is added to the current value of ACC. Program counter is 4. This part gets executed automatically after the loop value is entered.

TEMP = 5
ACC = 30 + 5 = 35
Loop Reg = 1

Program pointer 18: DISPACC

Here the value in accumulator is displayed in the output. Program counter is 5.

TEMP = 5
ACC = 35
Output = 35 Loop Reg = 1

Program pointer 19: ENDBLK

This block ends the loop when value in loop register becomes 0 or passes the program counter to 4 if loop register isn't 0. Program counter is 6.

TEMP = 5
ACC = 35
Output = 35 Loop Reg = 0 (Loop ends here)

Program pointer 20: HALT

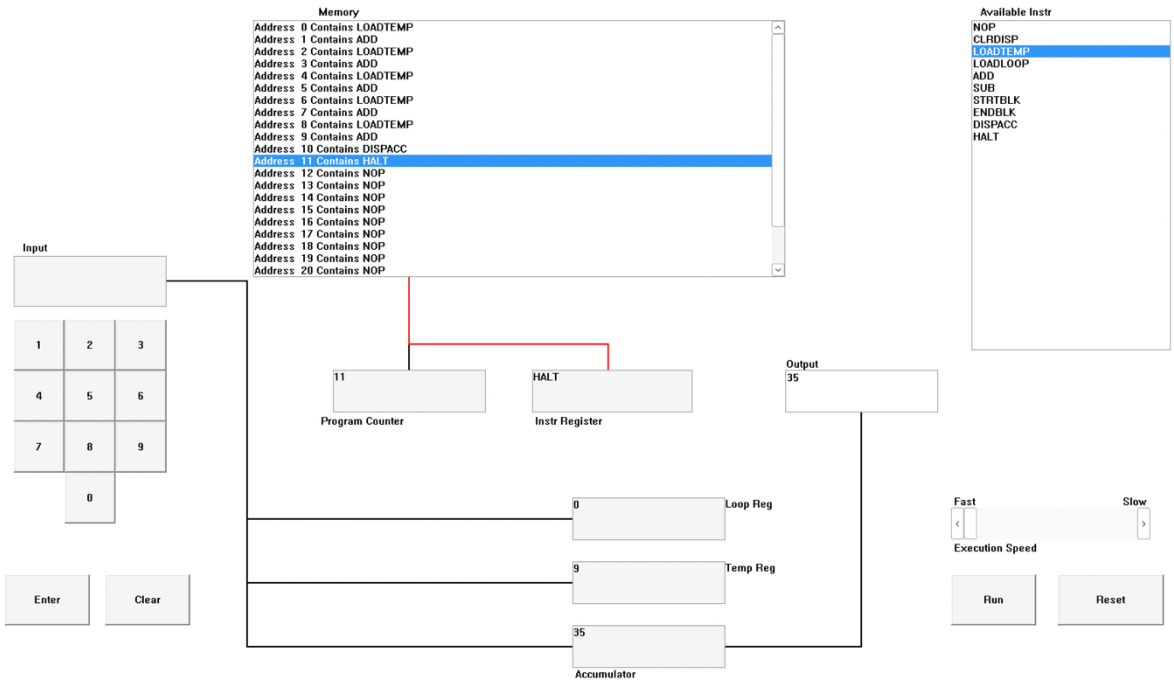
Here the program stops.

TEMP = 5
ACC = 35
Output = 35
Loop Reg = 0 (Loop ends here)
Program pointer = 7

e. **Write a program that will let you add five different value numbers together. List your program below and explain how it works (25 marks) (Make sure you include details of what is happening at every program pointer position to get full marks; you can include screen shots as well as descriptions to help illustrate your answer.)**

LOADTEMP
ADD
LOADTEMP
ADD
LOADTEMP
ADD
LOADTEMP
ADD
LOADTEMP
ADD
DISPACC
HALT

→ The above program loads five different values entered by user and adds them together. The program loads the number entered by user to the temporary register and then adds it the current value of accumulator. This process continues 5 times. And at the end the value in the accumulator is displayed to the output and the program stops.
Here,
LOADTEMP = 5,6,7,8,9



Program pointer 1: LOADTEMP

Here the value entered by user id loaded to the TEMP.
TEMP = 5
ACC = 0
Program counter = 0

Program pointer 2: ADD

Here the value in TEMP is added to the current value of ACC. This step is executed automatically after the value is entered by the user.
TEMP = 5
ACC = 0 + 5 = 5
Program counter = 1

Program pointer 3: LOADTEMP

Here the value entered by user id loaded to the TEMP.
TEMP = 6
ACC = 5
Program counter = 2

Program pointer 4: ADD

Here the value in TEMP is added to the current value of ACC. This step is executed automatically after the value is entered by the user.
TEMP = 6
ACC = 5 + 6 = 11
Program counter = 3

Program pointer 5: LOADTEMP

Here the value entered by user id loaded to the TEMP.
TEMP = 7
ACC = 11
Program counter = 4

Program pointer 6: ADD

Here the value in TEMP is added to the current value of ACC. This step is executed automatically after the value is entered by the user.
TEMP = 7
ACC = 11 + 7 = 18
Program counter = 5

Program pointer 7: LOADTEMP

Here the value entered by user id loaded to the TEMP.
TEMP = 8
ACC = 18
Program counter = 6

Program pointer 8: ADD

Here the value in TEMP is added to the current value of ACC. This step is executed automatically after the value is entered by the user.
TEMP = 8
ACC = 18 + 8 = 26
Program counter = 7

Program pointer 9: LOADTEMP

Here the value entered by user id loaded to the TEMP.
TEMP = 9
ACC = 26
Program counter = 8

Program pointer 10: ADD

Here the value in TEMP is added to the current value of ACC. This step is executed automatically after the value is entered by the user.
TEMP = 9
ACC = 26 + 9 = 35
Program counter = 9

Program pointer 11: DISPACC

Here the value of accumulator is displayed to the output.
TEMP = 9
ACC = 35
Output = 35
Program counter = 10

Program pointer 12: HALT

Here the program stops.
TEMP = 9
ACC = 35
Output = 35
Program counter = 11

f. Write a program that will let you multiply two fixed-value numbers together. List your program below and explain how it works (35 marks) (Make sure you include details of what is happening at every program pointer position to get full marks; you can include screen shots as well as descriptions to help illustrate your answer.)

Ans. The program that let multiply two fixed value numbers together is given below:

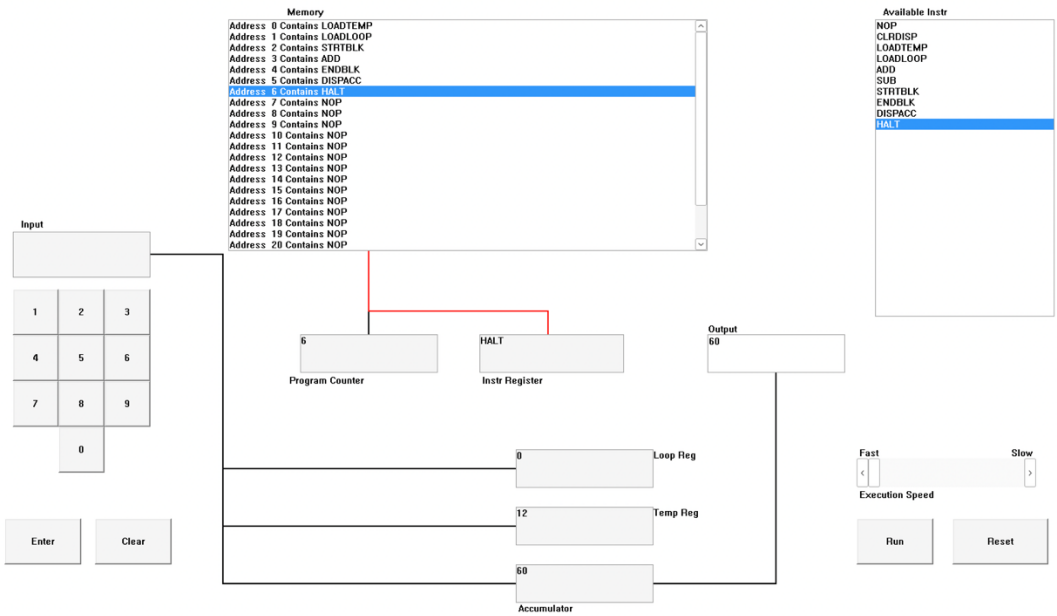
LOADTEMP
LOADLOOP
STRTBLK
ADD
ENDBLK
DISPACC
HALT

The above program gives the result of multiplication of any two fixed value numbers. It takes the number that we need to multiply and then it takes the number from which it is needed to be multiplied with. The logic used here is that the multiplicand means the number that is to be multiplied and multiplier means how many timed the number is to be added to itself.

LOADTEMP = multiplicand

LOADLOOP = multiplier

The program performs multiplication by repeatedly adding the same number using a loop. The analogy behind this program is that it takes a number and repeatedly adds the same number to the accumulator until the loop counter reaches zero. This process of repeated addition represents the fundamental meaning of multiplication, as multiplying a number is equivalent to adding that number to itself a specified number of times. The program uses a loop format for performing multiplication. Suppose we need to perform **12 * 6** which is equals to **60**. Below shown is the performed calculation:



The program takes,
LOADTEMP = 12
LOADLOOP = 6

Which means 12 is needed to be added to the accumulator 6 times to calculate the value of **12*6.** Below is the detailed step by step explanation of this calculation:
Temporary Register = TEMP
Accumulator = ACC

Program pointer 1: LOADTEMP

Here the value entered by the user is loaded to the TEMP. This value is multiplicand.

TEMP = 12
ACC = 0
Program counter = 0

Program pointer 2: LOADLOOP

Here the value entered by the user is loaded to the Loop Register. This value is multiplier. The value shows the number of times the loop iterates.

TEMP = 12
ACC = 0
Program counter = 1
Loop Reg = 5

(NOW THW LOOP ITERATES 5 TIMES AND ENDS WHEN THE VALUE OF LOOP REGISTER BECOMES 0)

Program pointer 3: STRTBLK

Here the block of loop starts.

TEMP = 12
ACC = 0
Program counter = 1
Loop Reg = 5

Program pointer 4: ADD

Here the current value of TEMP is added to the ACC. The process repeats 5 times so the TEMP is added to ACC 5 times i.e. 12+12+12+12+12

TEMP = 12
ACC = 12+12+12+12+12 = 60
Program counter = 3
Loop Reg = decreases at every **ENDBLK** and becomes 0 at the end of loop = 0

Program pointer 5: ENDBLK

Here the Loop ends as the value of Loop Register becomes 0. This process sends the counter to next instruction.

TEMP = 12
ACC = 60
Program counter = 4
Loop reg = 0
Output = 0

Program pointer 6: DISPACC

Here the current value of accumulator is displayed to the output. And the program counter is sent to next instruction.

TEMP = 12
ACC = 60
Program counter = 4
Loop reg = 0
Output = 60

Hence, the result of multiplication of two fixed number can be performed by using above program. This program shows how multiplication works by adding the same number again and again using a loop. One number is stored and the loop controls how many times it is added to the total. The accumulator keeps the running total until the loop finishes. When the program ends, the correct multiplied answer is shown. This proves that multiplication is repeated addition in a Von Neumann system.